

STM32L5 Series

Excellence in ultra-low-power with more security



First STM32 MCU based on Arm® Cortex®-M33 and TrustZone®

The STM32L5 series is the answer for embedded application requiring more security and a lower power consumption. It adds more security with Arm® Cortex®-M33 and its TrustZone® and ST security implementation while using the best-in-class ultra-low power technology.

Offering up to 512 Kbytes of flash (dual bank) memory and 256 Kbytes of SRAM, the STM32L5 reaches an upgraded level of performance (427 CoreMark) thanks to this new core and a new ST ART Accelerator™ (supporting now also external memory).

The STM32L5 offers a large portfolio with 7 packages (from 48 to 144 pins) and supports up to 125°C ambient temperature.

POWER CONSUMPTION

- EEMBC ULPBench®: 402 ULPMark-CP score
- Embedded SMPS step down converter (optional)
- Best power consumption numbers with full flexibility:
 - 33 nA in shutdown mode
 - 3.6 µA in stop mode with full SRAM and peripheral states retention and with 5 µs wake-up time
 - Down to 60 µA/MHz in active mode

A FULL SET OF SECURITY

- Flexible hardware and software secure isolations with TrustZone®
- Enhanced security services : dedicated secure user memory space for Secure Boot, symmetric and asymmetric crypto accelerations, memory and IP protection, independent Read out protection between secure / non secure domains, active IO tamper detection, certified cryptolib, embedded Secure Firmware Install loader and ecosystem.



More security



Lower power consumption



Integration, size performance



ST COMMUNITY

Engage with the community of STM32 enthusiasts on community.st.com/stm32

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